LECTURE NOTES

CRITICAL THINKING AND DEFINITIONS

(Think More, Think Better)

Primary Reference:
An Introduction to Critical Thinking and Creativity

1. Why Definitions Matter in Critical Thinking

Critical thinking begins with **clarity**, and clarity begins with **definitions**. To think critically, one must first understand **what exactly is being discussed**. Many disagreements in academic debates, legal cases, political arguments, and everyday conversations do **not arise from disagreement about facts**, but rather from **differences in how key terms are understood or used**.

People often assume they are disagreeing about reality when in fact they are disagreeing about **language**. This makes definitions foundational to reasoning, argument analysis, and decision-making.

Key idea

If we do not agree on what a term means, we cannot meaningfully argue about whether a statement involving that term is true or false.

For example, if two people argue about whether something is "fair," but one understands fairness as *equality* and the other as *equity*, their disagreement cannot be resolved without first clarifying the meaning of the term.

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Examples of definition-related disputes

Consider the following disputes:

- **Is an attack on two buildings one event or two events?**
  This question is not about what physically happened, but about how the word *event* or *occurrence* is defined.
- **Is a particular belief system a religion?**
  The answer depends on whether religion is defined in terms of belief in God, rituals, moral systems, social institutions, or something else.
- **Does a non-human animal have language?**
  The answer depends on how *language* is defined: communication in general, symbolic representation, grammar-based systems, or human-like speech.

In each case, **facts alone cannot resolve the issue**. The outcome depends crucially on **how the key terms are defined**.

2. Structure of a Definition

Most definitions follow a **simple logical structure**:

X is Y

Where:

- **X** is the term being defined
- **Y** is the defining expression

Example

A bachelor is an unmarried man.

Here:

- **Term:** "bachelor"
- **Defining phrase:** "unmarried man"

This structure allows us to clarify meaning by stating what **kind of thing** the term refers to and what **distinguishes it from other things**.

3. Terms, Referents, and Concepts

To understand how definitions work, it is essential to distinguish between **three closely related but distinct elements**: terms, referents, and concepts.

3.1 Term

A **term** is a **word or phrase** used in language to refer to something.

Examples:

- *water*
- *democracy*
- *United Nations*
- *inflation*

Terms are **linguistic items**. They exist in language, not in the world.

3.2 Referent

The **referent** is the **actual thing** (object, entity, property, or abstract idea) that a term refers to.

Examples:

- "Water" → the chemical substance **H_2O**
- "The United Nations" → an actual international organization
- "Inflation" → a real economic phenomenon involving general price increases

A referent exists **independently of language**, even though we use language to talk about it.

3.3 Concept

A **concept** is the **mental idea or understanding** associated with a term.

Concepts:

- Exist in the mind
- Are not made of words
- Can be shared across languages

Different languages may use **different terms** to express the **same concept**.

Example:

- English: *water*
- Spanish: *agua*

Here:

- Different **terms**
- Same **concept**
- Same **referent**

Understanding this distinction helps explain why translation is possible and why misunderstandings occur even when people use the same words.

4. Kinds (Types) of Definitions

In critical thinking, definitions are classified **by their function**, not merely by their wording.

Overview of the Three Main Types

Type of Definition	Purpose
Reportive	Reports existing usage
Stipulative	Assigns a new meaning
Precising	Reduces vagueness

Each type serves a **different reasoning purpose** and should be evaluated by **different standards**.

5. Reportive (Lexical) Definitions

5.1 What is a Reportive Definition?

A **reportive definition** (also called a **lexical definition**) attempts to **describe how a term is actually used** by a linguistic or professional community.

These are the kinds of definitions typically found in:

- Dictionaries
- Textbooks
- Encyclopedias

Examples

- *A prime number is a natural number greater than 1 that has exactly two distinct positive divisors.*
- *Sushi is a Japanese dish consisting of vinegared rice and other ingredients.*

These definitions aim to **report usage**, not to create new meanings.

5.2 Importance of Reportive Definitions

Reportive definitions are:

- **Essential for language learning**, especially in academic contexts
- **Central to disciplinary knowledge**, such as law, science, and economics
- **Necessary for shared understanding**, enabling effective communication

Without agreed reportive definitions, meaningful discussion becomes impossible.

5.3 Criteria for a Correct Reportive Definition

A good reportive definition must **accurately match how the term is actually used**.

(a) Avoid being too wide

A definition is **too wide** if it includes things that should not belong.

An airplane is a flying machine.

Problem:
This includes helicopters, drones, birds, and even flying toys, which are not airplanes.

(b) Avoid being too narrow

A definition is **too narrow** if it excludes things that clearly fall under the term.

Religion is belief in a creator god.

Problem:
This definition excludes non-theistic religions such as Buddhism and Jainism.

5.4 Dual Errors: Too Wide and Too Narrow

Some definitions fail in **both directions simultaneously**.

Example:

Bravery is not running away from danger.

Problems:

- **Too wide:** Includes reckless, foolish, or irresponsible behavior
- **Too narrow:** Excludes cautious or strategic bravery (e.g., firefighters assessing risk before acting)

This shows that crafting good definitions requires careful balance.

6. Practical Importance of Definitions: A Legal Case

Case Study: World Trade Center Insurance Dispute

After the September 11 attacks:

- Insurance contracts covered losses per **"occurrence"**
- The dispute centered on whether the attacks were **one occurrence or two occurrences**

The financial consequences were enormous:

- Billions of dollars depended on how one word was defined

Lesson

Definitions are not merely academic — they can have enormous legal, financial, and social consequences.

7. Stipulative Definitions

7.1 What is a Stipulative Definition?

A **stipulative definition** assigns a **new meaning** to a term, regardless of how it was previously used.

It does not aim to report usage, but to **introduce or fix meaning for a specific purpose**.

Examples

- In physics, a **quark** is defined as a type of elementary particle
- In economics, **risk premium** has a precise technical meaning

7.2 Key Characteristics

Stipulative definitions:

- Are **neither true nor false**
- Are judged by **usefulness and consistency**
- Are common in:
    - Science
    - Technology
    - Law
    - Theory-building

Once stipulated, the term **must be used consistently**.

7.3 Examples

- *Let "AI maturity" mean the level of organisational integration of artificial intelligence systems.*
- *By "digital poverty," we mean lack of access to both devices and digital skills.*

Such definitions enable **clear analysis and measurement**.

8. Precising Definitions

8.1 What is a Precising Definition?

A **precising definition** reduces **vagueness** in an existing term by setting **clear boundaries**.

It lies **between reportive and stipulative definitions**:

- It respects ordinary usage
- But sharpens it for practical decision-making

8.2 Examples

- A bus company defines *"old person"* as **anyone aged 65 or above**
- A university defines *"full-time student"* as **one enrolled in at least 60 credits per year**

These definitions:

- Do not claim to capture the "true meaning"
- Provide **clear operational rules**

8.3 Resolving Verbal Disputes

Many disputes are **verbal rather than factual**.

Classic example (William James):

- Hunters argue whether a man went *around* a squirrel
- Once "around" is clarified, the dispute disappears

Precising definitions help distinguish **real disagreements** from **mere word confusion**.

9. Criteria for Good Definitions

9.1 Use Intensional Definitions

Prefer defining by **essential characteristics**, not by listing examples.

✓ *A prime number is divisible only by 1 and itself.*
Listing all prime numbers

9.2 Avoid Circular Definitions

Time is what clocks measure.

Problem:

- Clocks are defined by measuring time

A definition should **not presuppose the term being defined**.

9.3 Ensure Clarity and Appropriate Precision

Avoid:

- Overly technical language
- Excessive precision that distorts meaning

A ship is a vessel longer than 30 meters.
This is arbitrary and misleading.

9.4 Avoid Obscure or Metaphorical Language

Science is searching for a black cat in a dark room.
This is poetic but not explanatory.

9.5 Avoid Persuasive Definitions

A **persuasive definition** sneaks emotional or evaluative content into a definition.

Example:

- Emotionally charged definitions of *democracy* or *abortion*

Critical thinking requires **neutral and fair definitions**.

10. Definition Techniques

10.1 Definition by Synonym

- *Physician = doctor*
- Simple but limited

10.2 Definition by Ostension

- Defining by pointing to examples
- Useful for colors or shapes
- Weak for abstract ideas

10.3 Definition by Genus and Differentia

Most powerful technique.

Structure:

X is a **genus** with **specific distinguishing features** (differentia)

Example:

A mule is an animal that is the offspring of a male donkey and a female horse.

11. Common Misconceptions About Definitions

- Not every word needs a precise definition
- Some concepts (e.g., *time*, *existence*) resist strict definition
- Dictionaries are starting points, not authorities
- Meanings evolve over time (avoid the **etymological fallacy**)

12. What We Have Learned

By the end of this lecture, you should understand:

- The **structure** of definitions
- The **three main types** of definitions
- **Criteria for good definitions**
- **Techniques** for defining terms
- Why definitions are central to **critical thinking, law, science, and innovation**

GROUP DISCUSSION QUESTIONS

(Critical Thinking and Definitions)

Instructions for Students

- Work in groups
- Assign roles: *facilitator, note-taker, presenter*
- Refer explicitly to **types of definitions** (reportive, stipulative, precising)
- Be prepared to justify your answers using **criteria for good definitions**

PART A: WARM-UP DISCUSSION

Question 1: Hidden Disagreements

Many disagreements are verbal rather than factual.

Discuss:

I. Can you think of a disagreement you have witnessed (academic, workplace, social media) that was actually about *definitions* rather than facts?
II. What key term was ambiguous?
III. How might clarifying the definition have resolved the disagreement?

HOMEWORK

To be submitted on or before nextweeks class

Identifying Types of Definitions

For each example below, identify whether the definition is **reportive, stipulative, or precising**, and explain WHY?

a) "A full-time employee is anyone who works at least 40 hours per week."
b) "By 'digital resilience,' we mean an organization's ability to recover from cyber disruption."
c) "A triangle is a three-sided polygon."

Evaluating Definitions

Consider the definition:

"Leadership is the ability to influence others."

Discuss:

I. Is this definition too wide, too narrow, or both?
II. What kinds of activities does it include that might not be leadership?
III. How could it be improved?

Persuasive Definitions

"A persuasive definition hides evaluation inside meaning."

Discuss:

I. Why are persuasive definitions problematic in critical thinking?
II. Can you think of a term in politics, business, or ethics that is often defined persuasively?
III. Should persuasive definitions ever be allowed?

Power of Words

Definitions shape decisions.

Discuss:

I. Why do definitions matter more in law, policy, and science than in everyday conversation?
II. Are there situations where vague definitions might actually be useful?